

Linear Algebra
Personal Notes

Mihir Rao

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1 The Geometry of Linear Equations

The basic computational problem of linear algebra is to solve a *system of linear equations*. A system with n equations and n unknowns can be understood in three complementary ways: one equation at a time, one column at a time, or as a single matrix equation.

1.1 A two-dimensional system

Consider

$$\begin{aligned}2x - y &= 0, \\ -x + 2y &= 3.\end{aligned}$$

In the *row picture*, each equation describes a line in the xy -plane. The first line passes through the origin and the point $(1, 2)$; the second contains $(-3, 0)$ and $(-1, 1)$. Their intersection is $(1, 2)$, so the simultaneous solution is

$$x = 1, \quad y = 2.$$

Substitution checks both equations. This picture makes the meaning of a solution immediate: it is a point satisfying every row at once.

The same system has a different geometric interpretation in the *column picture*. Grouping coefficients by unknown gives

$$x \begin{bmatrix} 2 \\ -1 \end{bmatrix} + y \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}.$$

A *linear combination* of vectors $c_1, \dots, c_k$ is an expression $a_1c_1 + \dots + a_kc_k$ with scalar coefficients a_i . Solving the system therefore means choosing the combination of the two coefficient columns that produces the right-hand side. Here

$$1 \begin{bmatrix} 2 \\ -1 \end{bmatrix} + 2 \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix},$$

which again yields $x = 1$ and $y = 2$.

1.2 The matrix equation

The compact algebraic form is

$$\underbrace{\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x \\ y \end{bmatrix}}_{\mathbf{x}} = \underbrace{\begin{bmatrix} 0 \\ 3 \end{bmatrix}}_{\mathbf{b}}, \quad A\mathbf{x} = \mathbf{b}.$$

The matrix A is the *coefficient matrix*, $\mathbf{x}$ is the vector of unknowns, and $\mathbf{b}$ is the right-hand-side vector.

The *matrix–vector product* $A\mathbf{x}$ should first be viewed as a combination of the columns of A . If

$$A = \begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix},$$

then

$$A\mathbf{x} = 1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 12 \\ 7 \end{bmatrix}.$$

Equivalently, each component is the dot product of a row of A with $\mathbf{x}$:

$$A\mathbf{x} = \begin{bmatrix} 2 \cdot 1 + 5 \cdot 2 \\ 1 \cdot 1 + 3 \cdot 2 \end{bmatrix}.$$

The row calculation and column combination are the same multiplication seen from different directions.

1.3 Three equations and three unknowns

For the system

$$\begin{aligned} 2x - y &= 0, \\ -x + 2y - z &= -1, \\ -3y + 4z &= 4, \end{aligned}$$

the matrix form is

$$\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -3 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix}.$$

In the row picture, each equation is a plane in three-dimensional space. Two planes generally meet along a line, and a third plane can reduce that common set to one point. That point is the solution. The idea is correct but the geometry quickly becomes difficult to draw in higher dimensions.

The column picture remains algebraically useful. It asks for scalars x, y, z such that

$$x \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} + y \begin{bmatrix} -1 \\ 2 \\ -3 \end{bmatrix} + z \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix}.$$

Because the right-hand side is the third column, the solution is immediately $\mathbf{x} = (0, 0, 1)^T$. A different right-hand side changes the required coefficients but not the columns being combined.

1.4 Which right-hand sides are reachable?

The equation $A\mathbf{x} = \mathbf{b}$ is solvable precisely when $\mathbf{b}$ is a linear combination of the columns of A . Thus the question

“Can we solve $A\mathbf{x} = \mathbf{b}$ for every $\mathbf{b}$?”

is equivalent to asking whether the column combinations fill the entire ambient space.

An n -by- n matrix is *invertible* (or *nonsingular*) when every right-hand side has exactly one solution. Geometrically, its columns provide enough independent directions to reach any vector. By contrast, columns are *linearly dependent* when one can be formed from the others. In three dimensions, if all three columns lie in one plane, every combination remains in that plane. Right-hand sides outside the plane are unreachable, and the matrix is *singular*.

This column viewpoint scales beyond what can be drawn: even for nine equations and nine unknowns, $A\mathbf{x}$ is still a linear combination of nine columns. The next task is to develop elimination, a systematic procedure for finding the coefficients when a solution exists.

Source: MIT OpenCourseWare, 18.06SC Linear Algebra (Fall 2011), Session 1, [The Geometry of Linear Equations](#), lecture by Gilbert Strang.